National University of Computer and Emerging Sciences, Lahore Campus

Course:	Electromagnetic Theory	Course Code:	EE305
Program:	BS(Electrical Engineering)	Semester:	Fall 2016
Duration:	60 Minutes	Total Marks:	50
Paper Date:	10-Nov-16	Weightage:	15%
Section:	ALL	Page(s):	1
Exam:	Midterm-2	Roll No:	

Instruction/Notes:

Question#1. [10]

Given a $90\mu C$ point charge located at the origin, find the total electric flux passing through a hemi-sphere $r = 26cm$. First plot the surface, guess the answer and then find the required flux.

Question#2. [15]

A sphere of radius 'a' has a uniform volume charge density of $\rho_o\ C/m^3$. Using Gauss's law, find the electric flux intensity inside and outside the sphere.

Question#3. [10+5=15]

(a) A total charge of $Q = \frac{40}{3} nC$ is uniformly distributed in the form of a circular disk of radius 2m. Find the potential due to this charge at a point on the axis, 2m from the disk.

(b) Again find the potential if same $Q$is located at the origin and compare it with **(a)**.

Question#4. [2+2+2+2+2=10]

The potential in free space is given by, $V = 2xy^2z^3 + 3\ln(x^2 + 2y^2 + 3z^2)$ V. Evaluate each of the following quantities at $P(3, 2, -1)$:

(a) V

(b) $\vec{E}$

(c) $\vec{D}$

(d) $\hat{a}_N$

(e) ρ_v